accenturetechnology

CLOUD ELEVATE

AWS CERTIFIED SOLUTIONS ARCHITECT

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Learning & Knowledge Management

Agenda

- AWS RDS
  - Introduction
  - Features
  - Benefits
- Aurora
  - Introduction
- NoSQL
  - Introduction
- DynamoDB
  - Introduction
  - Core Component
  - Backup
- DocumentDB
  - Introduction
- Keyspaces
  - Introduction
- Neptune
- QLDB
- Redshift
- Timestream

Relational Database

What is Relational Database ?

Amazon Relational Database Service (**RDS AWS**) is a web service that makes it easier to set up, operate, and scale a relational database in the cloud.
It provides cost-efficient, re-sizable capacity in an industry-standard relational database.

Database:

- Table
- Rows
- Columns(Field)

ID	First Name	Surname	Gender
1	Ryan	Kroonenburg	M
2	John	Adams	M
3	Julia	Clark	F
4	Danielle	Dustagheer	F

Relational Database

Relational Database on AWS:

- SQL Server
- Oracle
- MYSQL Server
- PostgreSQL
- Aurora
- MariaDB

Amazon RDS – Benefits

Easy to use : Amazon RDS database instances are pre-configured with parameters and settings appropriate for the engine and class you have selected.

Automatic software patching

Amazon RDS will make sure that the relational database software powering your deployment stays up-to-date with the latest patches.

Read Replicas

Read Replicas make it easy to elastically scale out beyond the capacity constraints of a single DB instance for read-heavy database workloads.

Cost-effectiveness

There is no up-front commitment with Amazon RDS; simply pay a monthly charge for each database instance that is launched.

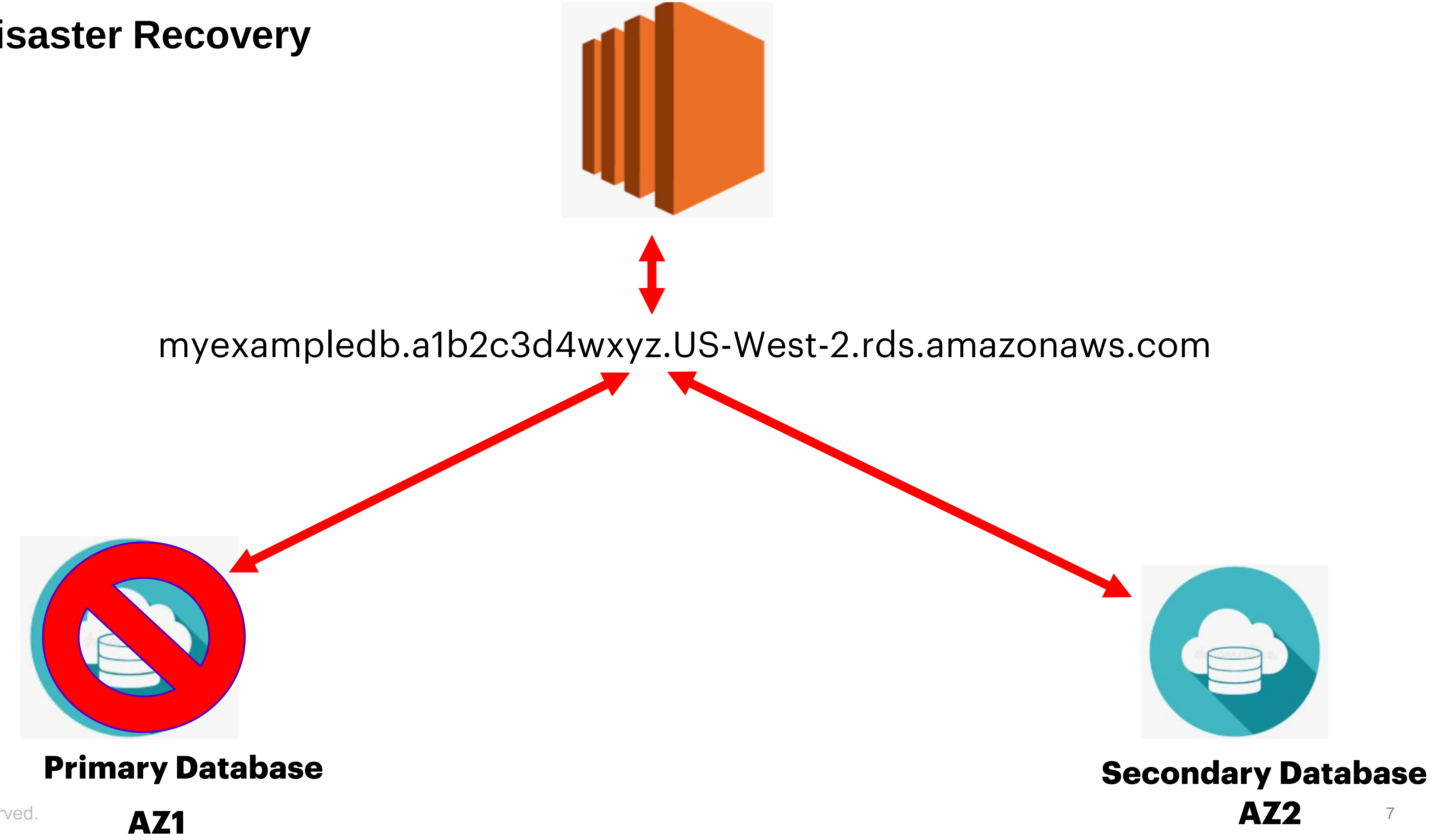
Relational Database

RDS has two key features:

- **Multi-AZ – For Disaster Recovery**
- **Read Replicas – For Performance**

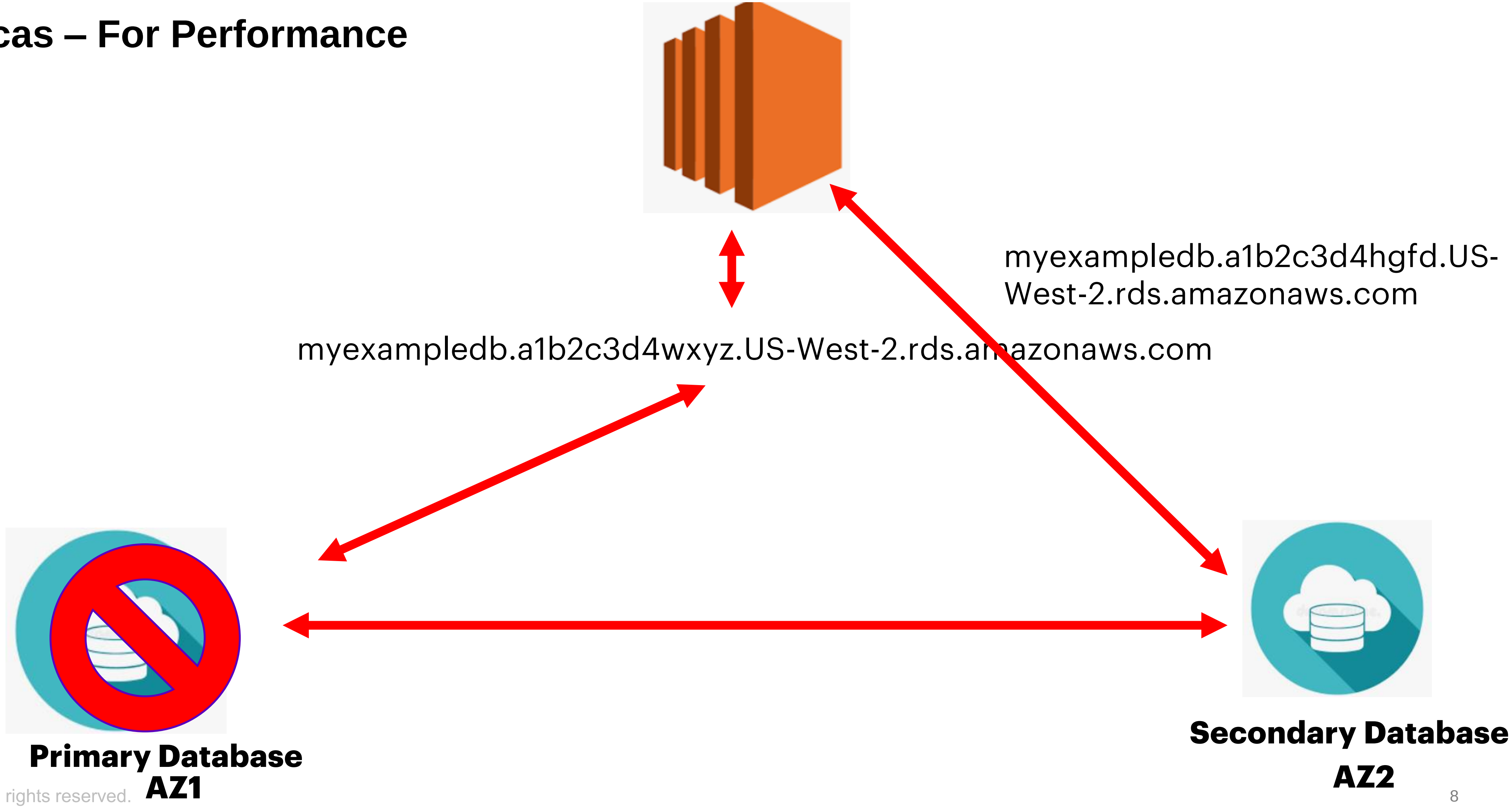
Relational Database

- **Multi-AZ – For Disaster Recovery**



Relational Database

- **Read Replicas – For Performance**



DEMO RDS

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Amazon Aurora

- Amazon Aurora is a relational database for the cloud which is a MySQL and PostgreSQL compatible database.
- With the ease of use and cost effectiveness, it combines the speed and availability of traditional enterprise databases
- Aurora delivers up to 5x performance of MySQL without requiring any changes to most MySQL applications
- Aurora PostgreSQL delivers up to 3 times the performance of PostgreSQL.
- It provides the security, availability, and reliability of commercial databases at the very low cost.

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Amazon Aurora Features:

- High Performance and Scalability
- High Availability and Durability
- Highly Secure
- Fully Managed
- Migration Support
- Cost-effectiveness
- Developer Productivity

Amazon Aurora Advantages

- Designed to offer 99.99% availability, replicating and maintaining 6 copies of your data across 3 Azs'.
- It is highly secured and uses SSL(AES-256),KMS and encryption and decryption to enhance security of database
- Automated backups are always enabled on AWS Aurora database instances.
- Aurora automatically fails over, if the primary instance in a database cluster fails, in the following order:
- →If Aurora Read Replicas are available, it promote an existing Read Replica to the new primary instance.

  →If no Read Replicas are available, then create a new primary instance.

INTRODUCTION TO NOSQL

- Provides flexible schemas for building modern applications
- Recognized for their ease of development, functionality, and performance at scale

Example: Relational Database

Table: Books

ISBN	Edition Number

Table: Authors

Author ID	Author Name

Table: Author-ISBN

Author ID	ISBN

Example: NoSQL

```
{
  ISBN: " ",
  Book Title:" ",
          Edition Number:" ",
                  Author Name: " ",
                          AuthorID:" "
                                 }
```

WHY NOSQL DATABASE?

Why should you use a NoSQL database?

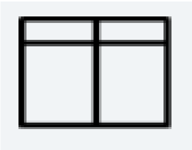
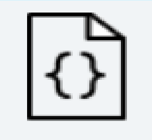
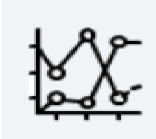
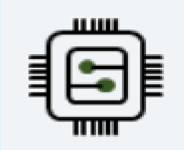
- NoSQL databases are good choice for applications such as mobile, web and gaming that requires scalable and flexible schemas

Benefits:

- Flexibility: Provide flexible schemas
- Scalability: Designed to scale out by using distributed clusters of hardware
- High-performance: NoSQL databases are optimized for specific data models and access patterns that provides higher performance
- Highly functional: Provide highly functional APIs and data types to build respective data models

NOSQL DATABASES

Types of NoSQL Databases

	Key-value: o It uses a simple key-value method to store data. A key-value database stores data as a collection of key-value pairs in which a key serves as a unique identifier. o Use Case: Gaming Application o Example: Amazon DynamoDB provides consistent single-digit millisecond latency for any scale of workloads
	Document: o Data is represented often as an object or JSON-like document o Use Case: Catalogs, user profiles, and content management systems o Example: Amazon DocumentDB (with MongoDB compatibility)
	Graph: o To build and run applications that work with highly connected datasets o Use Case: Social networking, recommendation engines, fraud detection o Example: Amazon Neptune
	In-memory: o It relies primarily on memory for data storage, in contrast to databases that store data on disk or SSDs o Use Case: Leader boards, session stores, and real-time analytics that require microsecond response o Example: Amazon ElastiCache offers Memcached and Redis and Amazon DynamoDB Accelerator (DAX)

AMAZON DYNAMODB

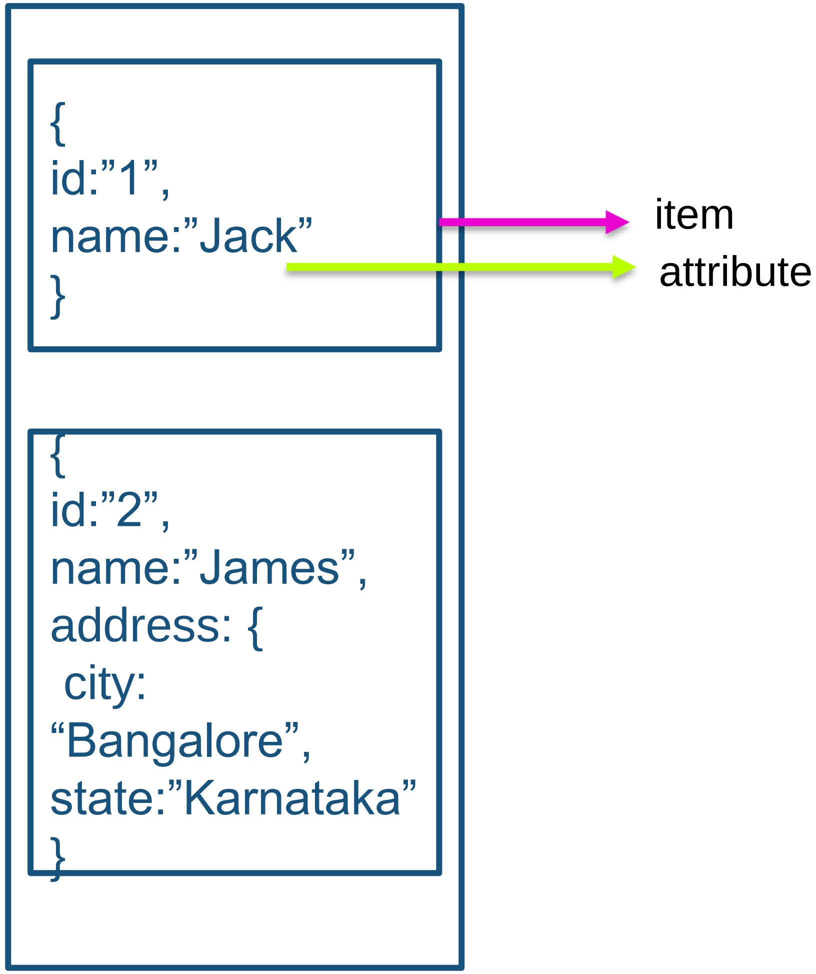
Amazon DynamoDB is a fast and flexible non-relational database service for all applications that need consistent, single-digit millisecond latency at any scale. It is a fully managed cloud database and supports both document and key-value store models.

Amazon DynamoDB

- Fully managed
- Fast, Consistent Performance
- Fine-grained access control
- Flexible

Amazon DynamoDB – Core components

- Tables
  - DynamoDB stores data in the form of tables like RDBMS
  - A table is a collection of data
  - DynamoDB tables are schema less (other than the primary key you need not define any other attributes)
- Items
  - A table can contain zero or more items
  - An item is a group of attributes that is uniquely identifiable among all of the other items
  - Item should be smaller then 400KB
- Attributes
  - Each item is composed of one or more attributes
  - Attributes are similar to fields or columns in other database systems



AMAZON DYNAMODB – CORE COMPONENTS

Example:

```
{
"Id":"123",
"Name":"sam",
"Age":"29"
}

{
"Id":"123",
"Name":["sam","Bob"],
}
```

Supported Datatypes:

- String
- Number
- Binary
- Boolean
- Null
- List (any types) - "List":[1,"sam",True]
- set of String(Unique values) "Friends":["Sam","bob"]
- set of Binary(unique values) "Images":["3d..affdd","4dffed...deff"]
- set of Number(unique value) "Number":[1,24,5,4,7,9]
- Map "Address": {
  "No":"123"
  "street":"xyz street"
  "city":"abc"
  }

Amazon DynamoDB – Backup

On-demand backup

- on-demand backup and restore capabilities allows you to create full backups of your tables.
- The backup is created asynchronously by applying all changes until the time of the request to the last full table snapshot

Point in time Recovery

- Automatic backup mechanism help to protect your Amazon DynamoDB tables from accidental write or delete operations
- DynamoDB maintains incremental backups of your table.
- restore that table to any point in time during the last 35 days.

DEMO DYNAMODB

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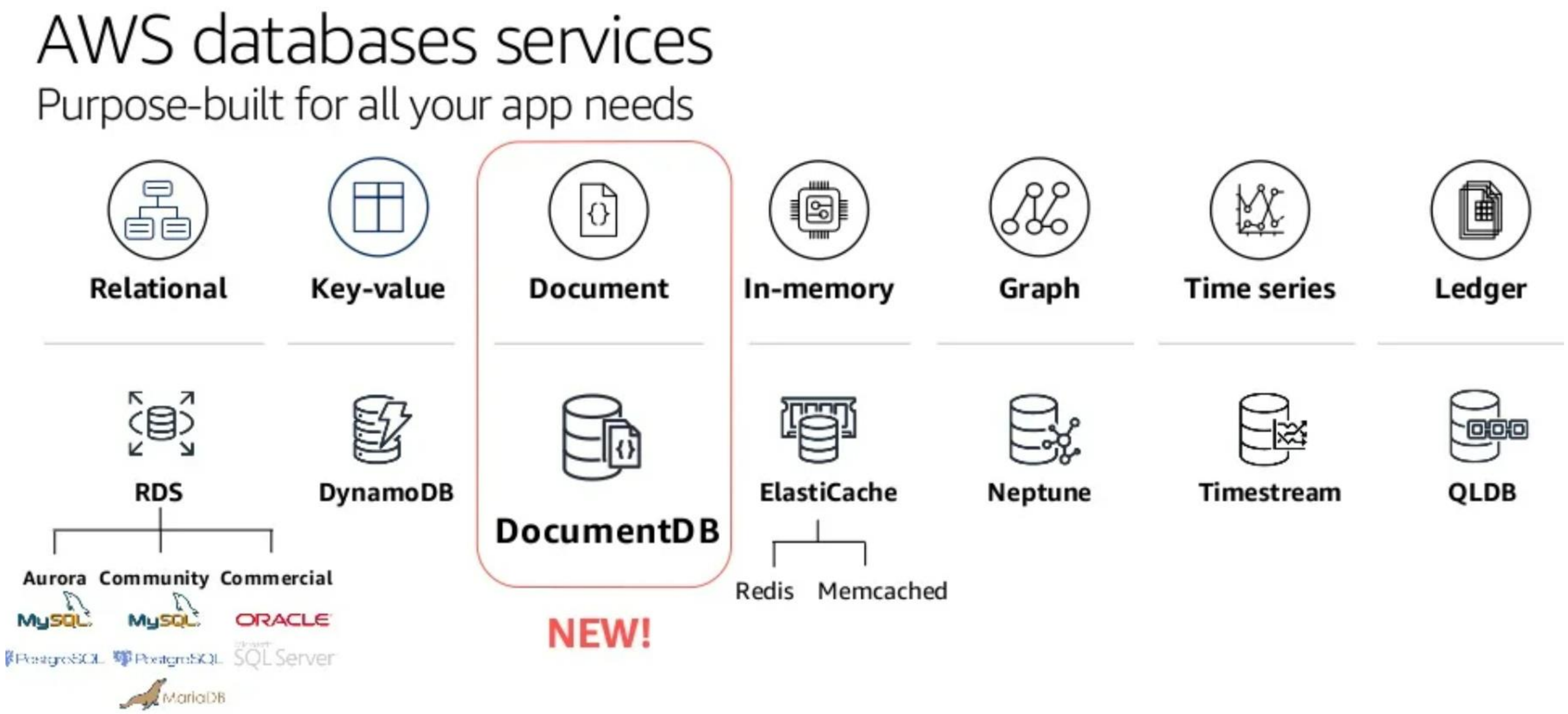
Amazon DocumentDB

Amazon DocumentDB (with MongoDB compatibility) features:



Amazon DocumentDB

Document databases

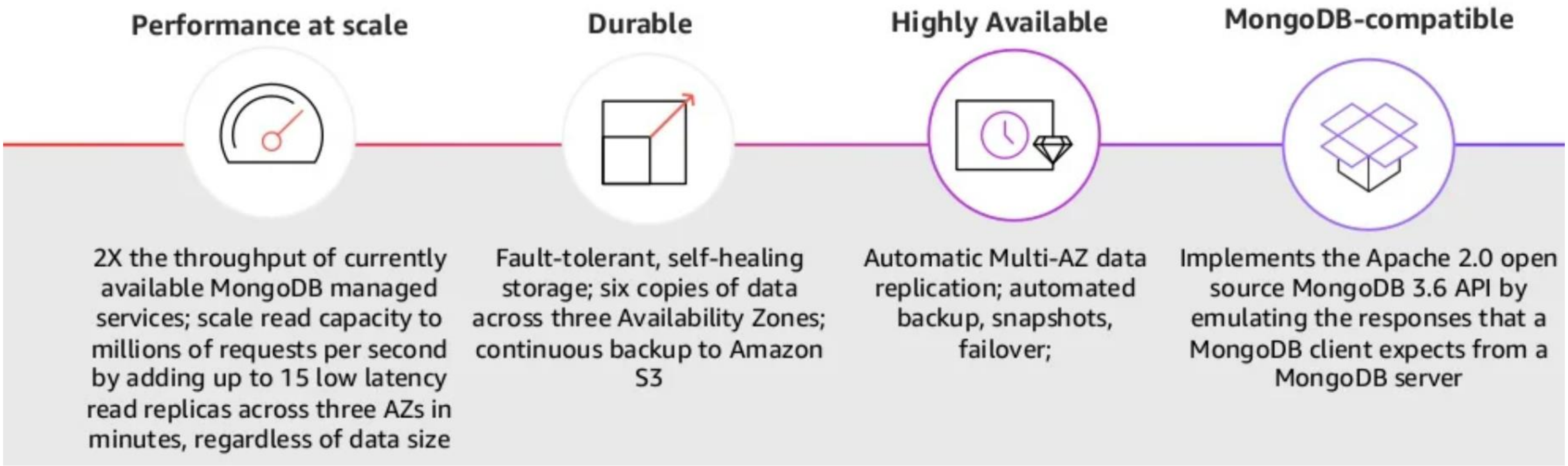
- Data is stored in JSON-like documents
- Documents map naturally to how humans model data
- Flexible schema and indexing
- Expressive query language built for documents (ad hoc queries and aggregations)

JSON documents are first-class objects of the database

```
{
  id: 1,
  name: "sue",
  age: 26,
  email: "sue@example.com",
  promotions: ["new user", "5%", "dog lover"],
  memberDate: 2018-2-22,
  shoppingCart: [
    {product:"abc", quantity:2, cost:19.99},
    {product:"edf", quantity:3, cost: 2.99}
  ]
}
```

Amazon DocumentDB

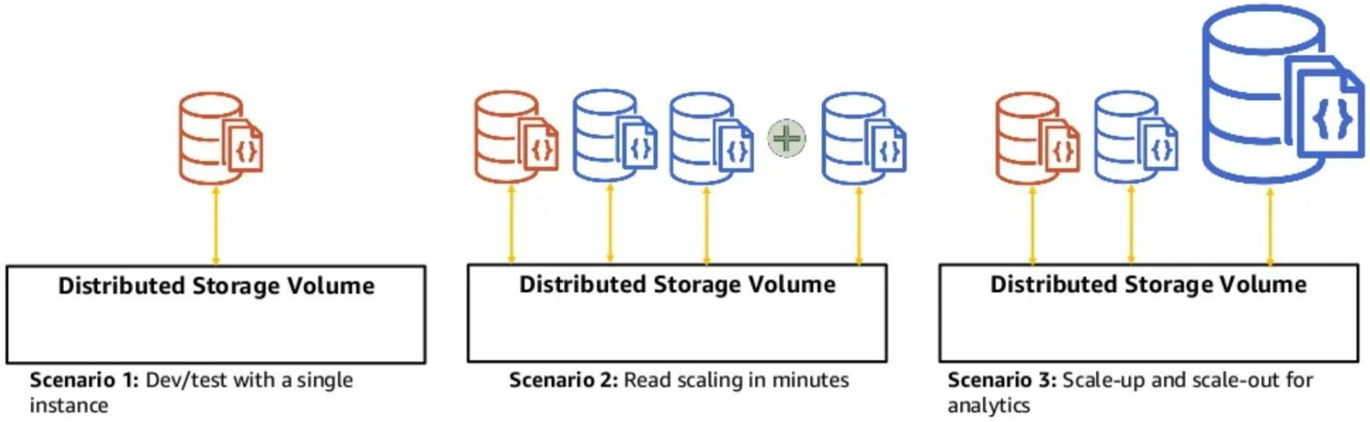
Fast, Scalable, highly available MongoDB compatible database:



Amazon DocumentDB

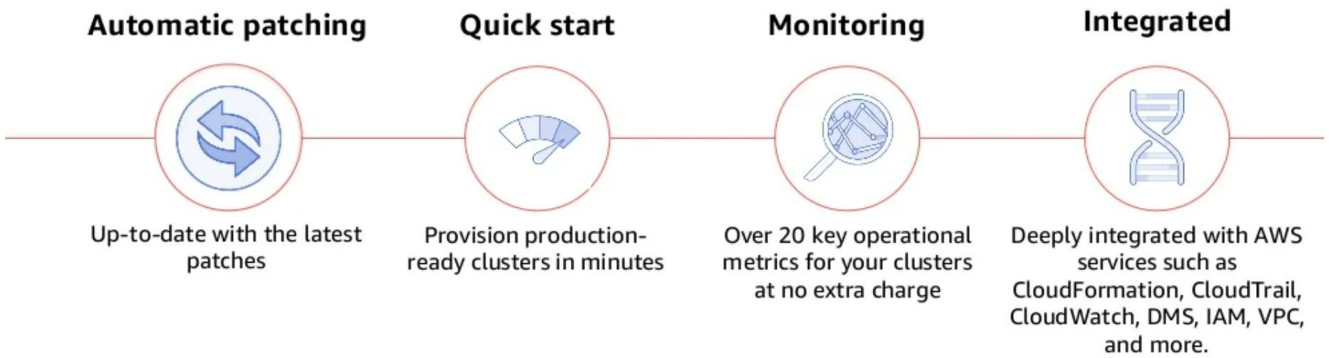
Flexible

Durability and replication are handled by the distributed storage volume



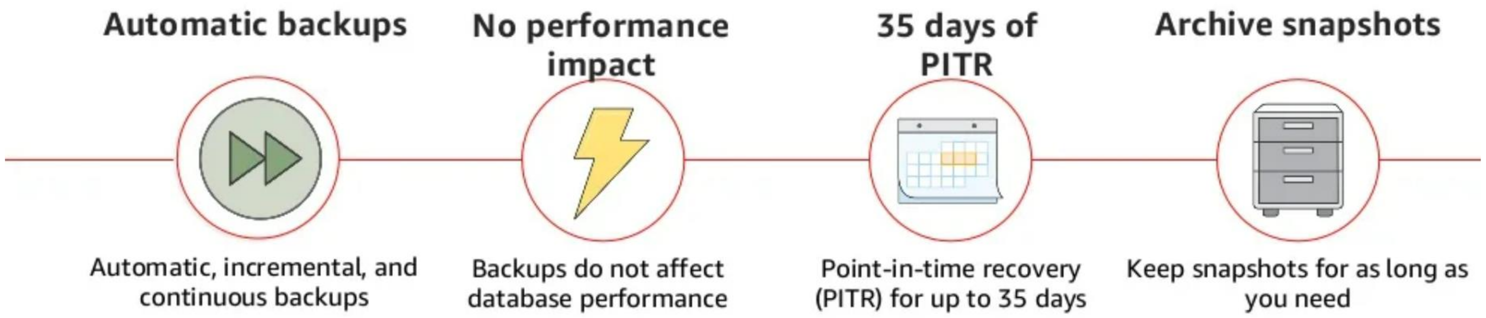
Fully-managed

Fast, reliable, and fully-managed MongoDB-compatible database service



Amazon DocumentDB

Backup



Amazon DocumentDB

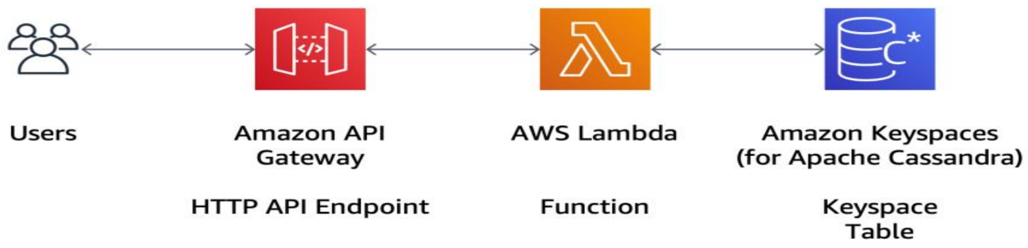
Which database to use?

DynamoDB	DocumentDB
• Key-value • Hyper Scale • Consistent low latency at any scale • Multi-master with global replication • Item size up to 400KB	• Documents • Efficiently query nested structures • Rich suite of query operators • Aggregations • Document size up to 16MB • Database size up to 64TB • Simple migration for existing MongoDB users

Amazon Keyspaces (for Apache Cassandra)

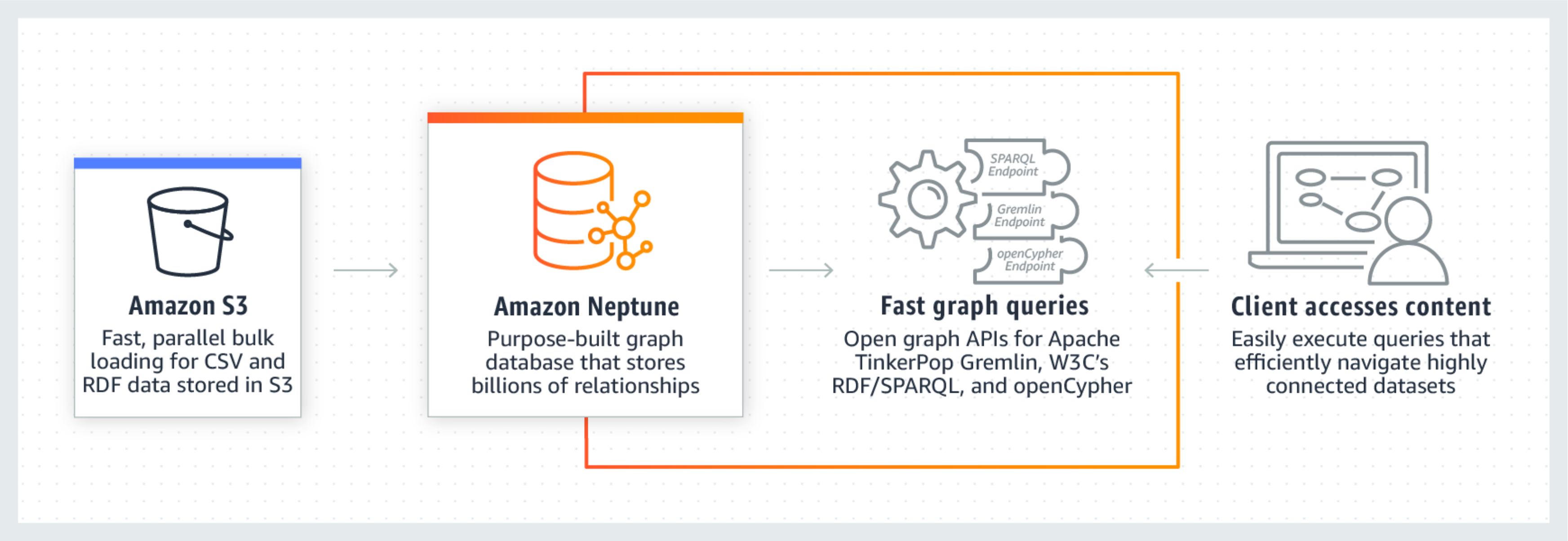
- Amazon Keyspaces (for Apache Cassandra) is a scalable, highly available, and managed Apache Cassandra–compatible database service.
- Amazon Keyspaces is built on Apache Cassandra, and you can use it as a fully managed, serverless database.
- Applications can read and write data from Amazon Keyspaces using existing Cassandra Query Language (CQL) code, with little or no changes.
- Amazon Keyspaces is serverless, pay for only the resources that you use and the service automatically scales tables up and down in response to application traffic.

  - AWS Lambda for the business logic.
  - Amazon API Gateway with the new HTTP API.
  - With Amazon Keyspaces, your data is stored in keyspaces and tables.
  - A keyspace gives you a way to group related tables together.



Amazon Neptune

Amazon Neptune is a fast, reliable, fully managed graph database service that makes it easy to build and run applications.

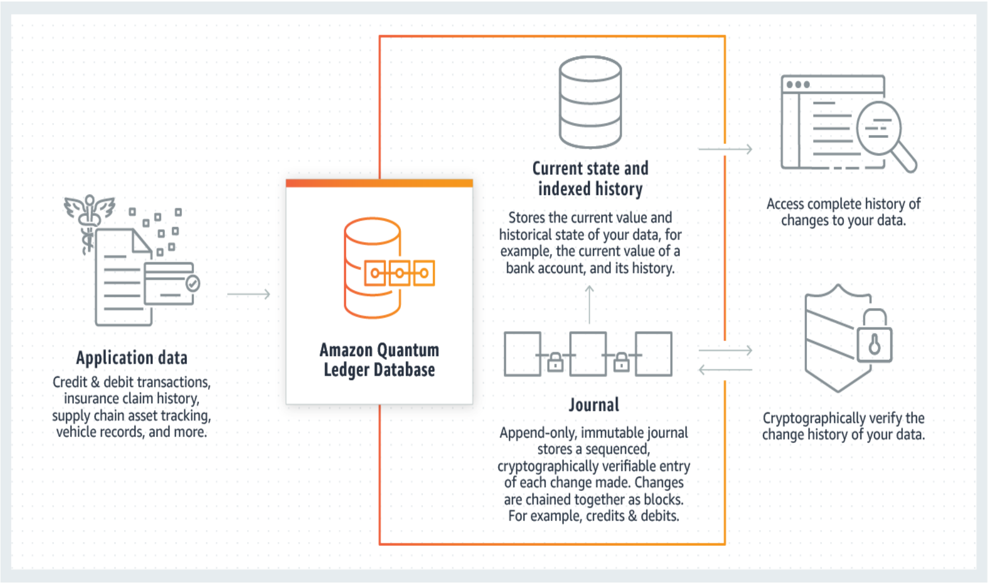


- Amazon Neptune, you can create sophisticated, interactive graph applications that can query billions of relationships in milliseconds.
- Amazon Neptune is a purpose-built, high-performance graph database engine.

Amazon QLDB

Amazon Quantum Ledger Database (QLDB) is a fully managed ledger database that provides a transparent, immutable, and cryptographically verifiable transaction log.

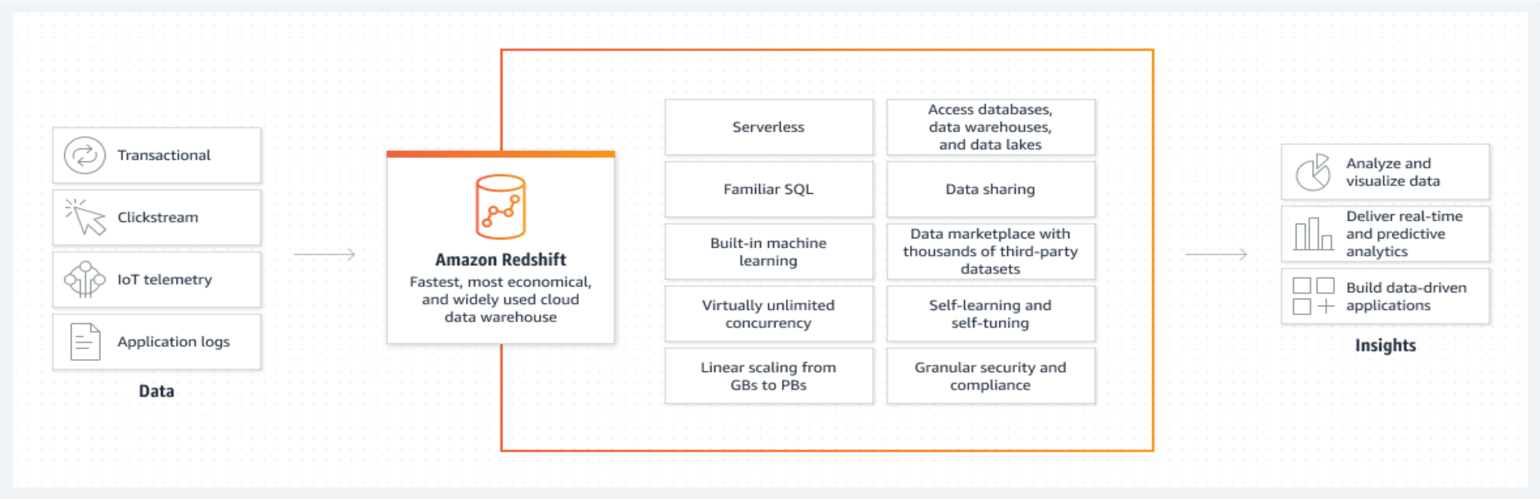
- **Store financial transactions**: Create a complete and accurate record of all financial transactions, such as credit and debit transactions.
- **Reconcile supply chain systems:** Record the history of each transaction and provide details of every batch manufactured, shipped, stored, and sold from facility to store.
- **Maintain claims history:** Track a claim over its lifetime, and cryptographically verify data integrity to make the application resilient against data entry errors and manipulation.
- **Centralize digital records:** Implement a system-of-record application to create a complete, centralized record of employee details such as payroll, bonus, and benefits.



Amazon Redshift

- Amazon Redshift is a fast and powerful, fully managed, petabyte-scale data warehouse service in the cloud.
- Amazon Redshift uses SQL to analyze structured and semi-structured data across data warehouses, operational databases, and data lakes
- using AWS-designed hardware and machine learning to deliver the best price performance at any scale.
- Amazon's Datawarehouse solution is called Redshift.

- Improve financial and demand forecasts
- Collaborate and share data
- Optimize your business intelligence
- Increase developer productivity



Amazon Redshift

Amazon Redshift Serverless:

- Amazon Redshift Serverless is a serverless option of Amazon Redshift that makes it easy to run and scale analytics in seconds without the need to set up and manage data warehouse infrastructure.
- With Redshift Serverless, any user—including data analysts, developers, business professionals, and data scientists—can get insights from data by simply loading and querying data in the data warehouse.

Amazon Redshift can be configured as follows:

- Single Node (160 GB)
- Multi-Node
    - Leader Node (manages client connections and receives queries)
    - Compute Node (store data and perform queries and computations. Up to 128 compute Nodes.

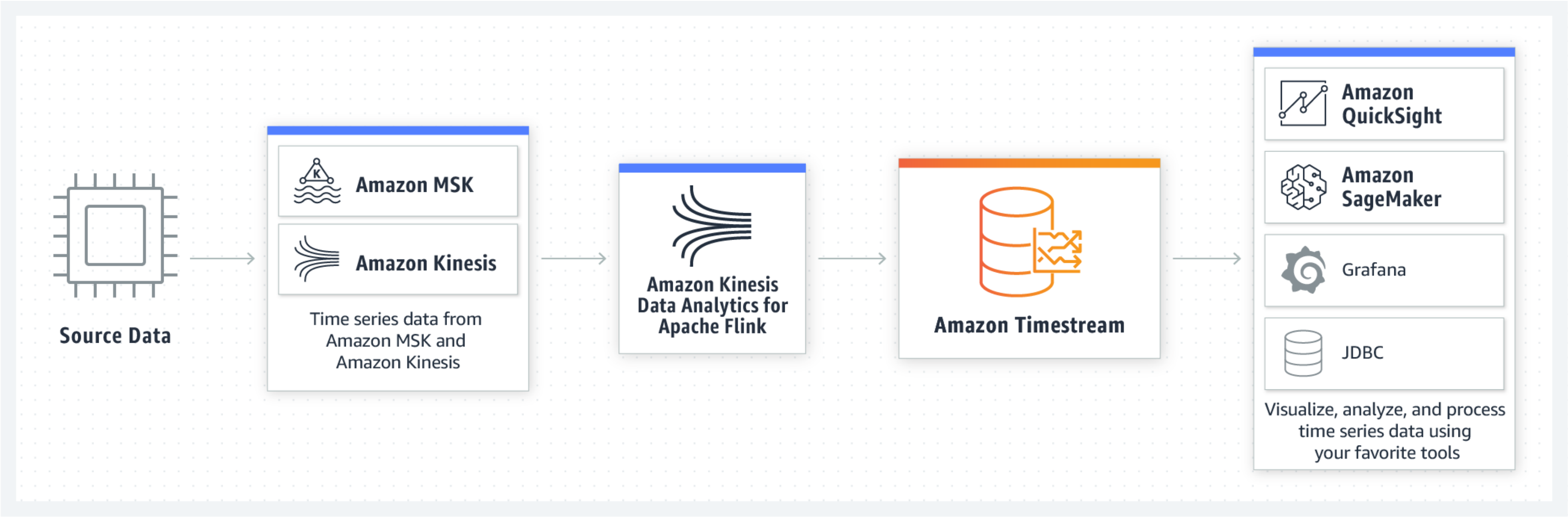
Amazon Timestream

- Amazon Timestream is a fully managed fast, scalable, and serverless time series database service for IoT and operational applications
- It makes easy to store and analyze trillions of events per day
- It is 1,000 times faster and at as little as 1/10th the cost of relational databases.
- Amazon Timestream has built-in time series analytics functions,
- It helps you identify trends and patterns in your data in near real-time.
- Amazon Timestream is serverless and automatically scales up or down to adjust capacity and performance,
- No need to manage the underlying infrastructure, freeing you to focus on building your applications.
    - High performance at low cost
    - Serverless with auto-scaling
    - Data lifecycle management

Amazon Timestream

Analytics applications

Amazon Timestream enables you to easily store and analyze data at scale.



Amazon Managed Streaming for Apache Kafka (Amazon MSK)

HAPPY LEARNING